CS215 Open Project Proposal

Team 20: Does Rain Make Onions Cry?

Studying the Effect of Heavy Monsoon Rainfall on Onion Prices in Indian Markets

1. Main Hypothesis / Research Question

Do unusually heavy monsoon rains increase the probability of sharp rises in onion prices in Indian agricultural markets? Excess rainfall may damage crops, disrupt transport, or delay arrivals, leading to sudden increases in market prices. However, onion prices also depend on storage, harvest cycles, and other supply conditions.

Our main null hypothesis (H_0) is that the probability of an onion-price spike is the same after unusually wet and normal-rainfall periods. The alternative (H_1) is that price spikes are more likely after unusually wet periods.

We model the price-spike indicator as a Bernoulli random variable conditional on rainfall conditions:

$$S \mid W = w \sim \text{Bernoulli}(p_w), \qquad p_w = P(S = 1 \mid W = w).$$

Our hypotheses are

$$H_0 : p_1 = p_0, \qquad H_1 : p_1 > p_0.$$

Alongside the probability of a spike, we will also study the *magnitude* of price changes. Let

$$\Delta_t = 100\frac{P_t - P_{t-1}}{P_{t-1}}$$

be the percentage change in the monthly representative onion price. As a working model, we will assume

$$\Delta_t \mid W = w \sim N(\mu_w, \sigma_w^2), \qquad w \in \{0, 1\},$$

where μ_1 and μ_0 are the mean percentage price changes in unusually wet and normal-rainfall periods, respectively. Our secondary hypotheses are

$$H_0^{(\mu)} : \mu_1 = \mu_0, \qquad H_1^{(\mu)} : \mu_1 > \mu_0.$$

The normal distribution is a working model for price changes, not a separate hypothesis to be formally tested. The Bernoulli model remains the primary analysis for the occurrence of price spikes; this continuous model separately examines the size of price changes.

Here, $W = 1$ denotes an unusually wet period and $S = 1$ denotes a sharp onion-price increase. Using the same price-change measure, we define the spike indicator as

$$S_t = \mathbf{1}\{\Delta_t \geq 20\%\}.$$

The 20% spike threshold is preliminary and will be checked using other thresholds. For rainfall, we will tentatively define $W = 1$ as at least 20% above the district-month normal and $W = 0$ as close to normal; unusually dry periods will be excluded from this comparison.

2. Source Papers, Reports and Other Reading

- AGMARKNET data portal and AGMARKNET training presentation: wholesale price data and reporting context.
- India Meteorological Department (IMD): rainfall observations, normals, and monsoon reports.
- Sekhar (2012), *Food Policy*: background on Indian agricultural markets.
- CEDA Agri-Market Data Portal: complementary agricultural market data.
- Further reading on onion-price volatility and monsoon-related supply disruptions will be selected after the feasibility check.

3. Datasets

- **Onion prices:** Daily mandi-level prices from AGMARKNET, with data.gov.in as an access option, aggregated to monthly prices.
- **Rainfall:** District-level monthly rainfall and historical normals from IMD, matched to market locations.
- **Validation:** Check coverage using CEDA and historical AGMARKNET extraction notes.
- **Scope:** Major onion-producing markets, initially in Maharashtra and nearby states, over the longest reliable period available (ideally 2018–2025).

4. Programming Languages / Tools

Python (pandas, SciPy, statsmodels, and matplotlib) and R will be used for data cleaning and analysis. Jupyter notebooks and GitHub will make the work reproducible.

5. Analysis Plan and Deliverables

1. **Data cleaning:** Check missing prices, duplicates, inconsistent units, extreme outliers, and rainfall coverage. Match mandis to districts and align the dates.
2. **Conditional probability:** Estimate $p_1 = P(S = 1 \mid W = 1)$ and $p_0 = P(S = 1 \mid W = 0)$. Report the estimates and large-sample confidence intervals using the normal approximation for sample proportions discussed in class.
3. **Conditional Bernoulli model (primary analysis):** Treat $S \mid W = w$ as Bernoulli(p_w) and formulate $H_0 : p_1 = p_0$ against $H_1 : p_1 > p_0$. The supplied slides cover inference for a single proportion but do not derive a formal test comparing two independent proportions, so the permitted two-group test will be confirmed with the teaching team.
4. **Continuous price-change model and hypothesis test (secondary analysis):** Retain the percentage price change Δ_t and use the working model $\Delta_t \mid W = w \sim N(\mu_w, \sigma_w^2)$ for $w \in \{0, 1\}$. Define $\theta = \mu_1 - \mu_0$ and estimate it by $\hat{\theta} = \bar{\Delta}_1 - \bar{\Delta}_0$. Test $H_0 : \theta = 0$ against $H_1 : \theta > 0$ using the general test-statistic and one-sided p-value framework from class. The exact standard error and degrees-of-freedom method for this two-group comparison will be confirmed with the teaching team.
5. **Distribution analysis:** Compare the spread of price changes, volatility, and unusually large increases using histograms and box plots. These plots will be descriptive checks, not formal tests of normality.
6. **Delayed effects and robustness:** Check rainfall in the same month and one or two previous months. Repeat the analysis with different spike thresholds and account for market and seasonal differences.

Main challenge: Rainfall may influence onion prices with a delay. Storage, harvest cycles, transportation, and other supply disruptions may also affect prices. Repeated observations from a market are not fully independent, so uncertainty estimates should account for clustering. The normal model is an approximation to be checked against the data. This study can identify associations, but not establish that rainfall alone causes price increases.

6. Logistics and Work Distribution

Team members will divide data collection, cleaning, exploratory analysis, statistical testing, and report writing. Code and progress updates will be maintained in the Team 20 GitHub repository. The team schedule spreadsheet will track tasks and deadlines.

7. LLM Use / Conversation Links

Any AI assistance used for drafting or coding will be documented according to course requirements.

8. Help Requested from the Teaching Team

We would appreciate guidance on defining unusually heavy rainfall, choosing a meaningful price-spike threshold, accounting for delayed effects, and obtaining reliable historical data from AGMARKNET and IMD. We would also appreciate confirmation of which formal procedures are acceptable for comparing two independent proportions and two group means, since the supplied slides present the general hypothesis-testing framework but do not derive these specific two-group tests.